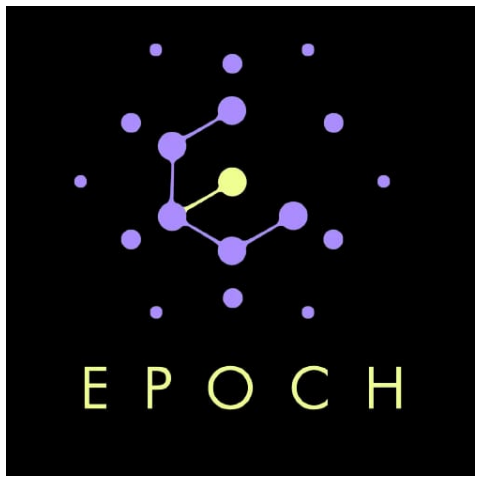
Mathematics for Machine Learning

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Probability

Probability

- Most of us would have come across the following formula for computing the probability of an event:

$$\text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

- But mathematically speaking, terms like "number of favorable outcomes" and "total number of possible outcomes" can be vague
- It is also unclear what an "event" actually is
- This is why we use **random variables** to formalize the above

Sample Space and Events

- A **random experiment** is a procedure that can be infinitely repeated and has a fixed, well-defined set of more than one possible outcomes
- Each repetition is known as a **trial**
- The above set of possible outcomes is known as the **sample space** of the experiment and is typically denoted by Ω
- An **event** is any subset of the sample space (can be the null set or even the complete set Ω itself)

$$A \subseteq \Omega$$

Random Variables

- To formalize the notion of an outcome of the trial, we assign a number to every possible outcome
- A **random variable** is a mapping (function) from the set of possible outcomes, i.e., the sample space, to the real numbers

$$X : \Omega \to \mathbb{R}$$

- Random variables are typically represented by uppercase letters and their instantiations are represented by lowercase letters, e.g., for some $\omega \in \Omega$

$$X(\omega) = x$$

Probability in terms of Random Variables

- Probabilities are associated with events. For example, consider the events $\{\omega : X(\omega) = 7\}$ and $\{\omega : X(\omega) \in \mathbb{N}\}$
- The probability associated with these events are denoted by

$$\Pr(\{\omega : X(\omega) = 7\}), \qquad \Pr(\{\omega : X(\omega) \in \mathbb{N}\}) \tag{1}$$

- But in most places, you will find that probabilities are represented in a much simpler format, like the following

$$\Pr(X = 7), \qquad \Pr(X \in \mathbb{N}) \tag{2}$$

- However, it is important to remember that this is just an abuse of notation. Always keep in mind that events are sets and random variables are functions

Example

- Suppose you have a regular six-sided dice. Your experiment consists of rolling it once and noting the number that appears on the top
- The sample space in this case is $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Now, suppose we are interested in the event where the number that appears on top of the dice is odd. The event can be mathematically written as $A = \{1, 3, 5\}$. Note that $A \subseteq \Omega$
- In this case, we can choose our random variable to be the square map $X : \{1, 2, 3, 4, 5, 6\} \to \{1, 4, 9, 16, 25, 36\}$
- The probability that the dice rolled an odd number is represented as

$$\Pr(X \in \{1, 9, 25\})$$

Random Vectors

- A **random vector** is simply a collection of several random variables defined on the same sample space

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$$

- Here, $X_i : \Omega \to \mathbb{R}$ for each $i = 1, \ldots, n$
- Random vectors, also known as **multivariate random variables**, are typically represented by boldface uppercase letters
- It is often useful to analyze data in multivariate form where each X_i corresponds to a different *feature* of the data

Elementary Set Theory

- **Union:** $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- **Intersection:** $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- **Complement:** $A^c = \{x : x \notin A\}$
- **Empty set:** $\phi = \{\}$
- **Subset:** $A \subseteq B$ if $x \in A \implies x \in B$
- **Proper subset:** $A \subset B$ if $A \subseteq B$ and $A \neq B$
- **Power set:** $\mathcal{P}(A) = \{X : X \subseteq A\}$
- **Inclusion-exclusion principle:** $|A \cup B| = |A| + |B| - |A \cap B|$
- **De Morgan's laws:**

$$(A \cup B)^c = A^c \cap B^c$$
$$(A \cap B)^c = A^c \cup B^c$$

Countability

- The number of elements in a set can be either finite or infinite
- However, all infinities are not the same
- A set A is said to be **countable** if and only if one of the following conditions is true:
  - $|A|$ is finite
  - $|A|$ is infinite but there is a bijective mapping $f : \mathbb{N} \to A$
- Are the following sets countable?
  1. $\mathbb{N}$
  2. $\mathbb{Z}$
  3. $\mathbb{Q}$
  4. $\mathbb{R}$

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  - $|A|$ is infinite but there is a bijective mapping $f : \mathbb{N} \to A$
- Are the following sets countable?
  1. $\mathbb{N}$ **YES**
  2. $\mathbb{Z}$ **YES**
  3. $\mathbb{Q}$ **YES**
  4. $\mathbb{R}$ **NO**
  5. $\mathbb{C}$ **NO**

Countability (contd.)

- A set being countable essentially means that the elements of the set can be "counted" in some order (one of which is given by $f(1), f(2), f(3), \ldots$)
- It can be thought of as a way to *index* the set
- The subset of a countable set is countable and the superset of an uncountable set is uncountable (Why?)
- The union of finitely many countable sets is countable (Why?)
- The Cartesian product of finitely many countable sets is countable (Why?)

Probability Axioms

- There are three axioms, known as **Kolmogorov axioms**, that form the foundation of probability theory
- Let Ω be the sample space and $A \subseteq \Omega$ be any event
  1. $\Pr(A) \geq 0 \quad \forall\ A \subseteq \Omega$
  2. $\Pr(\Omega) = 1$
  3. If $A_1, A_2, \ldots$ is a countable sequence of disjoint events, i.e., $A_i \cap A_j = \phi\ \forall i, j$ then

$$\Pr\left(\bigcup_i A_i\right) = \sum_i \Pr(A_i)$$

- The most interesting thing about these axioms is that *every* probability property can be derived from these three axioms

Probability Properties

Let us try to prove the following properties using the aforementioned axioms

1. If $A \subseteq B$ then $\Pr(A) \leq \Pr(B)$
2. $\Pr(\phi) = 0$
3. $\Pr(A^c) = 1 - \Pr(A)$
4. $0 \leq \Pr(A) \leq 1 \quad \forall\ A \subseteq \Omega$
5. $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$

Conditional Probability

- The idea of conditional probability is very useful in many cases
- The entirety of Bayesian learning is based on conditional probability
- The conditional probability of A given B where $A, B \subseteq \Omega$ is defined as

$$\Pr(A \mid B) \triangleq \frac{\Pr(A \cap B)}{\Pr(B)}$$

- For example, the conditional probability that the dice rolled 3 given that it rolled an odd number is represented by (considering the same random variable defined earlier)

$$\Pr(X = 9 \mid X \in \{1, 9, 25\})$$

Conditional Probability

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$$\Pr(A \mid B) \triangleq \frac{\Pr(A \cap B)}{\Pr(B)}$$

- What happens when $\Pr(B) = 0$?

Independence of Events

- Two events A and B where $A, B \in \Omega$ are said to be independent if and only if

$$\Pr(A \cap B) = \Pr(A) \Pr(B)$$

- Note that this also implies

$$\Pr(A \mid B) = \Pr(A)$$
$$\Pr(B \mid A) = \Pr(B)$$

- When is an event independent of itself?

Pairwise and Mutual Independence

- A countable sequence of events $A_1, A_2, \ldots$ is said to be pairwise independent if and only if

$$\Pr(A_i \cap A_j) = \Pr(A_i)\Pr(A_j) \quad \forall\ i, j$$

- A countable sequence of events $A_1, A_2, \ldots$ is said to be mutually independent if and only if

$$\Pr\left(\bigcap_i A_i\right) = \prod_i \Pr(A_i)$$

- Mutual independence implies pairwise independence
- Is the converse true?

Types of Random Variables

1. **Discrete Random Variable:** A random variable whose **support** is *countable*.
   e.g., number of heads in a series of coin tosses, number of people outside mess on a Sunday night, etc
2. **Continuous Random Variable:** A random variable whose **support** is *uncountable*.
   e.g., amount of rainfall in Sangareddy, calories burnt after gym, etc

Probability Mass Function

- A **probability mass function** (PMF) is the *probability distribution* defined for a discrete random variable
- It represents the probabilities associated with every value that the random variable can take
- Mathematically it is defined as

$$p_X : \mathbb{R} \to [0, 1]$$
$$p_X(x) = \Pr(X = x)$$

- The **support** of a random variable is defined as the set of values for which the PMF is non-zero

$$\operatorname{supp}(X) = \{x : p_X(x) > 0\}$$

Properties of the PMF

- Naturally, a probability mass function is always non-negative

$$p_X(x) \geq 0 \quad \forall\, x \in \mathbb{R}$$

- All the probability masses sum up to one

$$\sum_{x \in \mathbb{R}} p_X(x) = 1$$

- Therefore

$$0 \leq p_X(x) \leq 1 \quad \forall\, x \in \mathbb{R}$$

- The name *mass* comes from mass in Physics
- Just like physical mass, probability mass (the total probability) is conserved

Cumulative Distribution Function

- The **cumulative distribution function** (CDF) of a random variable represents the cumulative probability until a particular value
- It is defined for both discrete and continuous random variables
- Mathematically, for discrete random variables, it is defined as

$$
\begin{aligned}
F_X &: \mathbb{R} \to [0, 1] \\
F_X(x) &= \Pr(X \leq x) \\
&= \sum_{i \leq x} \Pr(X = i)
\end{aligned}
$$

- Note that the PMF can be uniquely determined from a given CDF (Why?)

Properties of the CDF

1. Every CDF is monotone increasing

$$x_1 < x_2 \implies F_X(x_1) \leq F_X(x_2)$$

2. Every CDF is right-continuous

$$F_X(x_0) = \lim_{x \to x_0^+} F(x)$$

3.

$$\lim_{x \to -\infty} F(x) = 0$$

4.

$$\lim_{x \to \infty} F(x) = 1$$

Expectation of a Discrete Random Variable

- The notion of expectation is a generalization of the weighted average
- The **expected value** (a.k.a. mean μ) of a random variable is the average value that one will observe after conducting many **independent** experiments
- Mathematically, the expectation of a discrete random variable is defined as

$$\mathbb{E}[X] \stackrel{\triangle}{=} \sum_{x \in \mathbb{R}} x \cdot p_X(x)$$

- Similarly, the expectation of a function of a random variable is defined as

$$\mathbb{E}[g(X)] \stackrel{\triangle}{=} \sum_{x \in \mathbb{R}} g(x) \cdot p_X(x)$$

Properties of the Expectation

1. If $X(w) \geq 0 \ \forall \ w \in \Omega$ then $\mathbb{E}[X] \geq 0$
2. **Linearity of expectation** (extremely important)

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y] \quad \forall \ a, b \in \mathbb{R}$$

   where X and Y are random variables defined on the same sample space
3. If $X(w) \geq Y(w) \ \forall \ w \in \Omega$ then $\mathbb{E}[X] \geq \mathbb{E}[Y]$
4. $\mathbb{E}[|X|] \geq |\mathbb{E}[X]|$

Expectation of a Random Vector

- Suppose we have the following random vector $\mathbf{X}$

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$$

- The expectation of this random vector is defined as

$$\mathbb{E}[\mathbf{X}] \triangleq \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{pmatrix}$$

Variance

- **Variance** of a random variable measures the average variation a random variable has from its mean
- Mathematically, the variance of a random variable is defined as

$$\mathrm{Var}\left[X\right] \triangleq \mathbb{E}\left[\left(X - \mathbb{E}\left[X\right]\right)^2\right]$$

- The square root of the variance is known as **standard deviation** or **standard error** and is denoted by σ

$$\sigma = \sqrt{\mathrm{Var}\left[X\right]}$$
$$\sigma^2 = \mathrm{Var}\left[X\right]$$

- What is the formula of variance for a discrete random variable?

Properties of the Variance

1. $\mathrm{Var}[X] \geq 0$ for any random variable X
2. $\mathrm{Var}[a] = 0$ where $a \in \mathbb{R}$ is some constant
3. $\mathrm{Var}[X + a] = \mathrm{Var}[X]$
4. $\mathrm{Var}[aX] = a^2 \mathrm{Var}[X]$
5. Note that $\mathrm{Var}[X + Y] \neq \mathrm{Var}[X] + \mathrm{Var}[Y]$ in general. If X and Y are *independent*, then the assertion is true
6. $\mathrm{Var}[X] = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2$

Discrete Random Variable

- Bernoulli Random Variable
- Multinouli Random Variable
- Binomial Random Variable
- Multinomial Random Variable

Bernoulli Random Variable

- Discrete Random Variable (X) whose support is $\{0, 1\}$ or the sample space contains only two possible outcomes ($|\Omega| = 2$) (basically a coin toss)
- If X is a Bernoulli Random Variable and let say it is used to associate the probability (p) of happening an event A, then we will map '1' for the success of the event A and map '0' for the failure of the event A. Mathematically,

$$X = \begin{cases} 1, & w.p. \quad p \\ 0, & w.p. \quad 1 - p \end{cases} \tag{3}$$

Bernoulli Random Variable

- And the PMF of X looks like,

$$p_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & x = 0 \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

- The expectation of bernouli random variable $E[X] = p$
- The variance of bernouli random variable $Var[X] = p(1 - p)$

Bernoulie Random Variable

Some Points :

- Bernoulli distribution only has a single parameter p
- A random experiment with exactly two possible outcomes is generally known as a **Bernoulli trial**
- It is not necessary that X should take only 0 or 1; instead any two discrete values can do the job but it is followed as part of tradition and mathematical convenience

Multinouli Random Variable

- Multinouli Random Variable : Discrete Random Variable (X) whose sample space has more than two possible outcomes ($|\Omega| > 2$).
  1. Generalised Bernoulli Random Variable.
  2. Support often taken as $\{1, ..., c\}$, if there are 'c' possible **outcomes**.
     Ex : A single roll of dice can give you a number in $\{1,2,3,4,5,6\}$ so here c $= 6$.
  3. From ML Perspective, if we are trying to predict a class (Y) of an object where the $|Y| > 2$ (Multi-Class Classification), then Y is represented by Multinouli Random Variable.

Multinouli Random Variable

- Similar to Bernoulli Random Variable, we can associate the probability of occurence (p_i) of each possible outcome $i(\in \{1, .., c\})$ and the PMF will be,

$$p_X(x) = \begin{cases} p_i & \forall x = 1, 2, \ldots, c \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

- As you can see the Mutinoulie "distribution" is parameterised with the set of probabilities $(p_1, p_2, \cdots, p_n)$.

Multinouli Random Variable

- The expectation of Multinouli Random Variable is,

$$E[X] = \sum_i i p_X(i)$$

- The variance of Multinouli Random Variable is,

$$Var[X] = \sum_i i^2 p_X(i) - \sum_i i p_X(i)$$

Binomial Random Variable

- It is a discrete random variable which counts the no .of times an event A occurs in a 'n' **independent** Bernoulli trails.
  Ex : Consider a random coin toss experiment (Bernoulli trail) which is repeated 'n' times independently, then one can observe count of no. of heads using a Binomial Random Variable (X).
- The Support of Binomial Random Variable would be {0,1,2,..,n} (Ex : No. of possible heads).
- **Note** : Don't confuse with Multinoulie Random Variable since the support looks similar, the support of Binomial Random Variable depend on n and represents the no .of times an event A occurs whereas in Multinoulie Random Variable support represents the category chosen.

Binomial Random Variable : PMF

- The PMF of Binomial Random Variable will be,

$$p_X(k) = \left(\frac{n!}{k!(n-k)!}\right)p^k(1-p)^{(n-k)} \tag{6}$$

  where k represents no. of times an event A occurred
  p is the probability associated to a Bernoulli Trail.
- The term $\left(\frac{n!}{k!(n-k)!}\right)$ refers to all possible combinations of getting 'k' successes out of 'n' independent Bernoulli trails.
- So, a Binomial Random Variable depends on no. of independent trails **n** and probability associated with each Bernoulli **p**.

Binomial Random Variable

- We can represent Binomial Random Variable as the sum of 'n' independent and identically distributed (i.i.d) Bernoulli Random Variables.
- Mathematically it can be written as,

$$X = X_1 + X_2 + \cdots + X_n \tag{7}$$

where X is a Binomial Random Variable and $\forall i, j$,

  - X_i and X_j are Independent.
  - And both X_i and X_j are from same Bernoulli **Distribution** (Identically Distributed).

Binomial Random Variable

- The expectation of Binomial Random Variable is $E[X] = np$
- The variance of Binomial Random Variable is $Var[X] = np(1 - p)$

Multinomial R.V

- Multinomial R.V : As the name suggests it is a generalisation of Binomial Random Variable where we observe specific **set of counts** of multiple categories.
- In place of Bernoulli here we will take independent **Multinoulie** trails.
- A Multinomial R.V similar to Binomial Random Variable, depends on no. of independent trails **n**.
- Then what will be the Support of Multinomial R.V (?).

Multinomial R.V

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- In place of Bernoulli here we will take independent **Multinoulie** trails.
- A Multinomial R.V similar to Binomial Random Variable, depends on no. of independent trails **n**.
- Then what will be the Support of Multinomial R.V (?). Basically the count of each category
- Is it a Random Variable or Random Vector (?)

Multinomial Random Vector

- Multinomial Random Vector : As the name suggests it is a generalisation of Binomial Random Variable where we observe specific **set of counts** of multiple categories.
- In place of Bernoulli here we will take independent **Multinoulie** trails.
- A Multinomial Random Variable similar to Binomial Random Variable, depends on no. of independent trails **n**.
- Then what will be the Support of Multinomial Random Variable (?).
  Basically the count of each category
- Is it a Random Variable or Random Vector (?) It is a Random Vector

Multinomial Random Vector with an example

- Consider 3 types of colored balls red(0), blue(1) and green(2) (Assume they are of large numbers). It is given that prob of picking a color i is p_i.
- Now we are interested in finding the probability of drawing 2 red balls, 2 green balls and 1 blue balls in 5 **independent** trails.
- As you can see whenever you draw a ball, you are picking a random color i from the three colors (Multinouli distribution) with $\Pr(X = i) = p_i$).
- In a particular case like below ,

RGRBG

Multinomial Random Vector Example Contd.

- The probability of drawing the balls in that sequence 02021 (2 reds, 2 greens,1 blue) will be

$$p_0 \times p_2 \times p_0 \times p_2 \times p_1 = p_0^2 p_2^2 p_1 \tag{8}$$

- Number of all possible sequences which result in 2 red balls, 2 green balls and 1 blue ball will be

$$\frac{5!}{2!2!1!} \tag{9}$$

- Overall probability of drawing 2 red balls, 2 green balls and 1 blue ball will then be given by $(\frac{5!}{2!2!1!})p_0^2 p_2^2 p_1$

Joint PMF OF Multinomial Random Vector

- Using the previous discussion, we can generalise each Multinoulie to 'c' categories and probability associated with each category 'i' being p_i.
- Then for **n** independent trails , the probability of getting a sequence $(n_1, n_2, \cdots, n_c)$ will be,

$$= \left(\frac{n!}{n_1! n_2! \cdots n_c!} \right) p_1^{n_1} p_2^{n_2} \cdots p_c^{n_c} \tag{10}$$

where, n_i represents the count of each category i with $\sum_{i=1}^{c} n_i = n$.

Joint PMF of Multinomial Random Vector

- The above expression is the **Joint PMF** of Multinomial Random Vector for a given count of each categories $X_1 = n_1, X_2 = n_2, \cdots, X_c = n_c$ where X_i is the Random Variable which represents number of occurrences of "particular" category i.
- Generally Joint PMF represented as,

$$p_{X_1,X_2,\cdots X_c}(n_1, n_2, \cdots, n_c) = \left(\frac{n!}{n_1! n_2! \cdots n_c!}\right) p_1^{n_1} p_2^{n_2} \cdots p_c^{n_c} \tag{11}$$

with Multinomial Random Vector $\mathbf{X} = (X_1, X_2, \cdots, X_c)^T$.

Multinomial Random Vector

- It is a multi variate version of binomial distribution.
- The expectation of Multinomial Random Vector,

$$E[\mathbf{X}] = \begin{pmatrix} np_1 & np_2 & \cdots & np_c \end{pmatrix}$$

- The variance of Multinomial Random Vector,

$$Var[\mathbf{X}] = \begin{pmatrix} np_1(1-p_1) & np_2(1-p_2) & \cdots & np_c(1-p_c) \end{pmatrix}$$

Exercise 1

Suppose you have a discrete random variable X taking values in $\mathcal{X} = \{0, 1, 2, \ldots, 99\}$ whose probability mass function is such that

$$p_X(x) \propto x^2 \quad \forall\, x \in \mathcal{X}$$

1. Plot the PMF of X
2. Plot the CDF of X

Cumulative Distribution Function

- Cumulative distribution function of a continuous random variable is still defined as

$$F_X : \mathbb{R} \to [0, 1]$$
$$F_X(x) = \Pr(X \leq x)$$

- It satisfies all the four properties of CDF from before
- It is also worth noting that any function that satisfies the four properties is the CDF of some random variable

Probability Density Function

- **Probability density function** (PDF) is the continuous analog of the probability mass function
- But the most important thing about the PDF is that **it is not equal to the probability associated with a value like PMF**
- PDF got its name from *density* in Physics and represents exactly that
- When the probability *density* f_X is integrated over some *volume* $\mathrm{d}x$, we obtain the probability *mass*

$$\int_a^b f_X(x)\,\mathrm{d}x = \Pr\left(a \leq X \leq b\right)$$

Probability Density Function (contd.)

- Another important thing to note that is **the probability of a single point is always zero in case of continuous random variables**

$$\Pr(X = a) = 0 \quad \forall\ a \in \mathbb{R}$$

- This is because a single point occupies *zero volume* and thus results in zero mass

$$\Pr(X = a) = \Pr(a \leq X \leq a) = \int_a^a f_X(x)\ \mathrm{d}x = 0$$

- It should be pretty clear by now that $f_X(a) \neq \Pr(X = a)$ unlike PMF

Probability Density Function (contd.)

$$
\begin{aligned}
F_X(x) &= \Pr(X \leq x) \\
&= \int_{-\infty}^{x} f_X(x)\, \mathrm{d}x \\
\therefore f_X(x) &= \frac{\mathrm{d}}{\mathrm{d}x} F_x(x) \\
&= \lim_{h \to 0} \frac{F_X(x+h) - F_X(x)}{h} \\
&= \lim_{h \to 0} \frac{\Pr(x < X \leq x+h)}{h}
\end{aligned}
$$

This result can be interpreted as "mass per unit length" in Physics

Properties of the PDF

1. For any continuous random variable X

$$f_X(x) \geq 0 \ \forall \ x \in \mathbb{R}$$

2. The area under the entire density curve is 1

$$\int_{\infty}^{\infty} f_X(x) \, \mathrm{d}x = 1$$

3. However, $f_X(x)$ is not upper bounded by 1 or any number because it is not a probability
4. In fact, there are distributions where the density not even finite, e.g., Dirac delta

Expectation of a Continuous Random Variable

- The expectation of a continuous random variable is defined as

$$\mathbb{E}[X] \triangleq \int_{-\infty}^{\infty} x \cdot f_X(x)\, \mathrm{d}x$$

- Similarly, the expectation of a function of a continuous random variable is defined as

$$\mathbb{E}[g(X)] \triangleq \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, \mathrm{d}x$$

Continuous Random Variable

- Uniform Random Variable
- Gaussian Random Variable
- Multivariate Gaussian

Uniform Random Variable

- A continuous random variable X is said to have Uniform Distribution in $[a, b]$ if its PDF $f_X(x)$ is

$$f_X(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

- As the name suggests, the random variable X takes all the values in interval $[a, b]$ with equal probability
- The expectation of the uniform random variable $E[X]$ is ,

$$E[X] = \frac{a+b}{2}$$

Uniform Random Variable

- The variance of the uniform random variable $Var[X]$,

$$Var[X] = \frac{(b-a)^2}{12}$$

- The CDF of X will be

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & x \in [a, b] \\ 1 & x > b \end{cases}$$

Gaussian Random Variable

- The noise/error in the measurements are generally modelled using Gaussian random variables
- Thanks to the Central Limit Theorem, it plays a crucial role in **learning** from data
- It has many real life applications

Gaussian Random Variable

- A continuous random variable (X) is said to have Gaussian/Normal Distribution i.e., $X \sim \mathcal{N}(\mu, \sigma^2)$ if its PDF $f_X(x)$ is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right) \quad x \in \mathbb{R}$$

Here, μ is the mean of Gaussian $E[X]$
σ is the standard deviation of the X

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  Here, μ is the mean of Gaussian $E[X]$
  σ is the standard deviation of the X

- In the special case when Normal distribution has $\mu = 0$ and $\sigma = 1$ we call it as **Standard Normal**.

Gaussian Random Variable

- The PDF of normal random variable is a bell-shaped curve and it is symmetric about mean μ and σ signifies the spread of the bell curve.

Gaussian Random Variable

- And we can relate any Normal random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ with standard normal Z using the following relationship,

$$X = \sigma Z + \mu$$

Gaussian Random Variable

- And we can relate any Normal random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ with standard normal Z using the following relationship,

$$X = \sigma Z + \mu$$

- The general result will be *linear combination of any n Normal random variables will be a Normal Random Variable* and has the following properties,

$$X = c_1 X_1 + c_2 X_2 + \cdots + c_n X_n$$

$$E[X] = c_1\mu_1 + c_2\mu_2 + \cdots + c_n\mu_n$$
$$Var[X] = c_1^2\sigma_1^2 + c_2^2\sigma_2^2 + \cdots + c_n^2\sigma_n^2$$

Gaussian Random Variable

- The CDF of the standard normal Z will be,

$$F_Z(z) = \Pr(Z \leq z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} exp\left(\frac{-z^2}{2}\right)$$

- $F_Z(z) + F_Z(-z) = 1$
- $F_Z(0) = \frac{1}{2}$

Multivariate Gaussian

- Generalisation of the Gaussian random variable
- From the perspective of machine learning, if all the *features* in the data takes continuous values then entire data can be considered as a **multivariate Gaussian**

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$$

Multivariate Gaussian

- We can say that the random vector $\mathbf{X}$ is multivariate Gaussian if the individual components are **jointly Gaussian** i.e., any linear combination of the individual components should be a Gaussian.
- A Multivariate Gaussian $\mathbf{X}$ is represented as $\mathbf{X} \sim (\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- $\boldsymbol{\mu}$ is the $E[\mathbf{X}]$

Multivariate Gaussian

- And Σ is the **covariance matrix** of $\mathbf{X}$ i.e.,

$$\Sigma = E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)^T]$$

Multivariate Gaussian

- And Σ is the **covariance matrix** of $\mathbf{X}$ i.e.,

$$\Sigma = E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)^T]$$

- Then the joint density,

$$f_{X_1, X_2, \cdots X_n}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{X} - \mu)^T \Sigma^{-1} (\mathbf{X} - \mu)\right)$$

Multivariate Gaussian

- The special case of Multivariate Gaussian is when each pair of (X_i, X_j) are independent.
- It is like assuming each *feature* is independent to each other. This is known as **Naïve Bayes** Asssumption since it is *naïve*.
- In that case the Joint density looks like,

$$f_{X_1,X_2,\cdots X_n}(\mathbf{x}) = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

Exercise 2

Suppose you have a continuous random variable X taking values in $\mathcal{X} = [0, \infty)$ whose probability density function function is such that

$$f_X(x) = \begin{cases} \alpha e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

This distribution is known as the **exponential distribution**

1. Find the value of α such that the above is a valid PDF
2. Plot the PDF of X
3. Plot the CDF of X

Joint Distribution

- It is often the case that we are interested in more than one random variable
- **Joint distribution** is a distribution associated with a random vector (of the same type - either all discrete or all continuous)
- If $X_1, \ldots, X_n$ are discrete random variables then the joint PMF of $X_1, \ldots, X_n$ is defined as

$$p_{X_1 \ldots X_n} : \mathbb{R}^n \to [0, 1]$$
$$p_{X_1 \ldots X_n}(x_1, \ldots, x_n) = \Pr(X_1 = x_1, \ldots, X_n = x_n)$$

- Note that $\Pr(X = x, Y = y)$ denotes the probability of the intersection of the events corresponding to $X = x$ and $Y = y$

Joint PMF

- Since Joint PMF represents probabilities

$$0 \leq p_{XY}(x, y) \leq 1 \quad \forall\, x, y \in \mathbb{R}$$

- Joint PMF must also sum up to one (notice the double summation)

$$\sum_x \sum_y p_{XY}(x, y) = 1$$

- The expectation of a function of multiple discrete random variables is defined using the joint distribution

$$\mathbb{E}\left[g(X, Y)\right] \triangleq \sum_x \sum_y g(x, y) \cdot p_{XY}(x, y)$$

Joint CDF

- If $X_1, \ldots, X_n$ are random variables then the joint CDF of $X_1, \ldots, X_n$ is defined as

$$F_{X_1 \ldots X_n} : \mathbb{R}^n \to [0, 1]$$
$$F_{X_1 \ldots X_n}(x_1, \ldots, x_n) = \Pr\left(X_1 \leq x_1, \ldots, X_n \leq x_n\right)$$

- If additionally, the random vector is continuous, i.e., $X_1, \ldots, X_n$ are continuous random variables, then the joint PDF is defined as

$$f_{X_1 \ldots X_n} : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$$
$$f_{X_1 \ldots X_n}(x_1, \ldots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{X_1 \ldots X_n}(x_1, \ldots, x_n)$$

Joint PDF

- Just like last time

$$\int_a^b \int_c^d f_{XY}(x, y)\,\mathrm{d}x\,\mathrm{d}y = \Pr\left(a \leq X \leq b, c \leq Y \leq d\right)$$

- Also, we have

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y)\,\mathrm{d}x\,\mathrm{d}y = 1$$

- The expectation of a function of multiple continuous random variables is given by

$$\mathbb{E}\left[g(X, Y)\right] \triangleq \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) \cdot p_{XY}(x, y)\,\mathrm{d}x\,\mathrm{d}y$$

Marginal

- The marginals of a joint are the individual components involved in the join
- Say F_{XY} is a joint distribution, then F_X and F_Y are said to be the **marginal distributions** of F_{XY}
- The methods (known as **marginalization**) to compute the marginals from the joints of different types are as follows:

$$F_X(x) = \lim_{y \to \infty} F_{XY}(x, y)$$

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, \mathrm{d}y$$

Conditional Distribution

- If X and Y are both discrete random variables, then the conditional PMF of X given Y is defined as

$$p_{Y|X}(y \mid x) = \frac{p_{XY}(xy)}{p_X(x)}$$

- If X and Y are both continuous random variables, then the conditional PDF of X given Y is defined as

$$f_{Y|X}(y \mid x) = \frac{f_{XY}(xy)}{f_X(x)}$$

- When one of them is discrete and the other is continuous, the conditional distribution is still defined even though the joint is undefined

Independence of Random Variables

- Two random variables X and Y are said to be independent if and only if

$$F_{XY}(x, y) = F_X(x)\, F_Y(y)$$

or equivalently

$$\begin{aligned} p_{XY}(x, y) &= p_X(x)\, p_Y(y) && \text{for discrete} \\ f_{XY}(x, y) &= f_X(x)\, f_Y(y) && \text{for continuous} \end{aligned}$$

- If X and Y are independent random variables, then

$$\mathbb{E}\left[g(X)h(Y)\right] = \mathbb{E}\left[g(X)\right]\mathbb{E}\left[h(Y)\right]$$

but note that $\operatorname{Var}\left[XY\right] = \operatorname{Var}\left[X\right]\operatorname{Var}\left[Y\right]$ is not true in general even if they are independent

Bayes' Rule for Events

- Bayes' rule is one of the most fundamental theorems in probability
- If $A, B \subseteq \Omega$ are events, then Bayes' rule states that

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\ \Pr(A)}{\Pr(B)}$$

Bayes' Rule for Events

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- Proving Bayes' rule for events is trivial as it follows directly from the definition of conditional probability
- A much more non-trivial variant of the Bayes' rule is presented in the next slide

Bayes' Rule for Random Variables

- If X and Y are random variables of any combination of types **(both discrete, or both continuous, or one discrete one continuous)** then we have

$$p^*(X \mid Y) = \frac{p^*(Y \mid X)p^*(X)}{p^*(Y)}$$

  where p^* denotes the respective distribution based on the type of the random variable

- The beauty of this result is that this holds true even though the join is not defined when one random variable is discrete and other is continuous and even though $\Pr(Y = y) = 0$ when Y is continuous
- All the terms in this theorem are well-defined

Total Expectation Rule

- If X and Y are random variables of any combination of types **(both discrete, or both continuous, or one discrete one continuous)** then the total expectation rule states that

$$E[g(X,Y)] = E\left[E\left[g(X,Y) \mid X\right]\right] = E\left[E\left[g(X,Y) \mid Y\right]\right]$$

where g(X,Y) is any function of X and Y

Exercise 3

Suppose that the expected number of accidents per week on the highway through IIT-H is **4**. Suppose also that the numbers of commuters injured in each accident are independent random variables with a common mean of **2**. Assume also that the number of commuters injured in each accident is independent of the number of accidents that occur. What is the expected number of injuries during a week on that highway? Compute this formally using the *total expectation rule*.

Statistics

Variable Types

Categorical variables are variables that represent characteristics or attributes that are not numerical in nature.

- Nominal variables: Nominal variables are categorical variables that do not have an inherent order or ranking.
- Ordinal variables: Ordinal variables are categorical variables that have a natural order or ranking.

Quantitative variables are those that represent numerical values.

- Continuous variables: Continuous variables are quantitative variables that can take on any value within a certain range.
- Discrete variables: Discrete variables are quantitative variables that can only take on specific values..

Measures of Central Tendency

The statistical values that provide information about the center of a distribution or dataset.

- **Mean:** The mean is the average value of a dataset. Mathematically expressed as $\frac{\sum_{i=1}^{i=n} N_i}{n}$.
- **Median:** The median is the middle value of a dataset when the values are arranged in order.
- **Mode:** The mode is the value that occurs most frequently in a dataset.

Sampling of data

The two major types of sampling would be:-

- **Simple Random Sampling**: In this method, every member of a population has an equal chance of being selected.
- **Stratified Sampling**: This method divides the population into subgroups and then randomly selects samples from each group based on proportionate representation.

One dimensional Plots

One Dimensional data, basically an array of numeric values or categorical values. There are multiple ways to visualize 1-D data, the main ones being:

- **Histogram**:- A histogram is a bar graph-like representation of data that shows the distribution of values along a single axis.
- **Boxplot** :- A boxplot is a graphical representation of the distribution of data based on five summary statistics: minimum, first and third quartile, median, and maximum.

Two dimensional Plots

These plots are a type of data visualisation where the points are displayed along two axes. Some of the important 2D plots are :-

- **Scatter plot**:- A scatter plot is a plot that shows the relationship between two variables by displaying the data points as dots on a two-dimensional coordinate system.
- **Line plot**:- To display the variation of a variable with respect to another quantitiy.Lineplots have lots of importance in data involving any series or especially data with timeseries.

Priori and Posteriori

- **Priori** is the parameters or information that one can assume or one knows before any data or dataset is analyzed.
- **Posteriori** is the information that one can infer after analyzing the data/ performing the experiment.

Covariance

- **Covariance** is defined as a measure to determine how much two different variables correlate.
- If an increase in one variable leads to an increase in another variable, likewise for the decrease in one variable leading to a decrease in the other one, we call it as **Positive Covariance**.
- In the other case where an increase in one variable leads to a decrease in the other, then we call it as **Negative Covariance**
- If the two variables are completely independent, they will have zero variance.

Cross Entropy

It is a very commonly used loss function used in Machine learning Classification tasks. It is used to predict the similarity of the predicted probability distribution to the actual probability distribution.
Mathematically expressed by

$$\sum_{i=1}^{i=n}(p_i * log(q_i))$$

where there are "n" classes, with **p** being the true probability distribution and **q** being the predicted probability distribution.

Estimation of Means

- For a collection of N values, having mean μ, we take a sample of these values each containing n numbers. We then find the mean of the the sample values which we take it as X and the variance being s.
- We can say that as n tends to N, X will also tend to μ.
- Likewise if we take a repeated number of samples, each with the same size (n), and then plot the distribution of means, it would result in a normal distribution, with the centre being at μ and the standard deviation equal to $\sigma/\sqrt{n}$

Errors

- Good model prediction must contain fewer errors
- For a Function, f denotes the true relationship between x and y. y is defined as

$$y = f(x) + \epsilon$$

where ϵ is the error of the model. This can be defined as

- $\epsilon = \text{bias}(\hat{f}(\text{x}))^2 + \text{var}(\hat{f}(\text{x})) +$ Irreducible error.
- Here, Bias and Variance are the reducible errors

Bias

- The inability of a model to accurately capture the true relationship is called **bias**.
- **High bias Vs Low bias**
- Mathematically, it is defined as

$$bias[\hat{f}(x)] = E[\hat{f}(x)] - f(x)$$

Variance

- **Variance** is the amount by which a different training data set would alter the estimate of the true relationship.
- **High Variance Vs Low Variance**
- Mathematically, it is defined as the expectation of the squared deviation of $\hat{f}(x)$ from its expected value $\mathrm{E}[\hat{f}(x)]$.

$$var(\hat{f}(x)) = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

The Trade off

- Over-fitting Vs Under-fitting
- Accurate Model: Low bias and Low variance

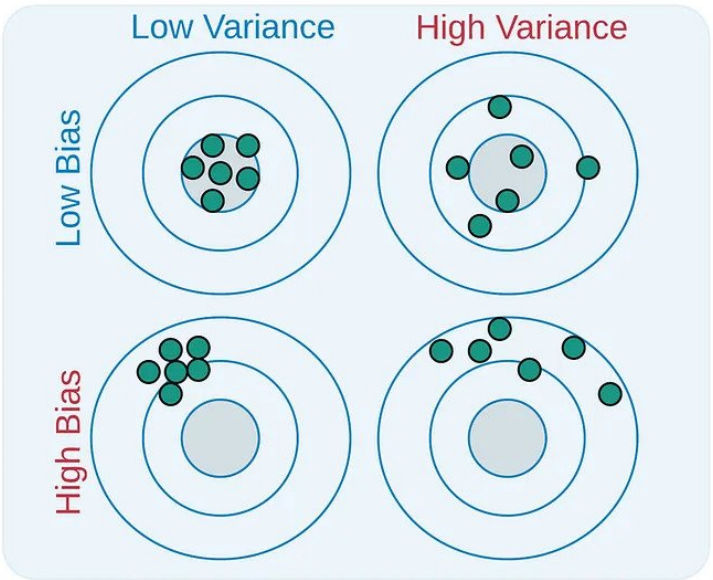


Figure: Tradeoff

Confusion Matrix

- A **confusion matrix** is a table that defines the classification algorithm's performance.

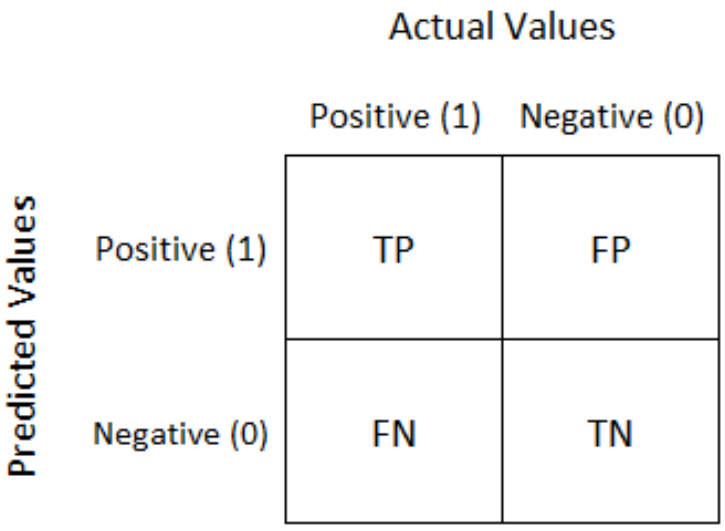


Figure: Confusion Matrix

Precision and Recall

- Precision is the ratio between the True Positives and all the Positives.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

- Recall is the measure of our model correctly identifying True Positives.

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

Exercise!

Let's say you want to predict how many people are infected with a contagious virus in times before they show the symptoms and isolate them from the healthy ones. The two values for our target variable would be Sick(1) and Not Sick(0). Draw the Confusion Matrix and find out the Precision and Recall

ID	Actual Sick?	Predicted Sick?
1	1	1
2	0	0
3	0	0
4	1	1
5	0	0
6	0	0
7	1	0
8	0	1
9	0	0
10	1	0

Statistical Inference

- Statistical inference is a method of making decisions about the parameters of a population, based on random sampling.
- Helps to assess the relationship between the dependent and independent variables.
- The difference between descriptive and inferential statistics lies in the process as much as it does the statistics that you report.

Law of Large Numbers

- The Law of Large Numbers tells us that as the sample size increases and goes to ∞, the probability that $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i = \mu$ approaches 1.
- Let $X_1, X_2, X_3 \ldots$ be the random variables with finite expected value $\mathrm{E}[X_i] = \mu < \infty$,
  Then for any $\epsilon > 0$

$$\lim_{x \to \infty} P(|\overline{X} - \mu| \geq \epsilon) = 0$$

Central Limit Theorem

- Let $X_1, X_2, X_3 ...$ be the random variables with finite expected value $E[X_i] = \mu < \infty$ and $0 < Var(X) < \infty$. Then, the random variable,

$$Z_n = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{X_1 + X_2 + X_3 + ... + X_n - n\mu}{\sqrt{n}\sigma}$$

converges in distribution to the standard normal random variable as n goes to infinity, that is

$$\lim_{x \to \infty} p(Z_n \leq x) = \Phi(x)$$

for all $x \in \mathbb{R}$, where $\Phi(x)$ is the standard normal CDF.

The Pollster's Problem

The Boss asks pollster to calculate the number of people to be randomly picked as a sample of population to estimate the value of the fraction of population that will vote "yes" in a referendum (denoted by 'p')

Pollster's Initial Approach and its Limitations

Steps

- 'n' people are picked randomly, uniformly and independently over the population and their answers are recorded in indicator random variables

$$\Pr(X_i) = \begin{cases} 1, & \text{if yes} \\ 0, & \text{if no} \end{cases} \tag{12}$$

- Fraction of "yes" in our sample $= M_n = \frac{X_1 + X_2 + \cdots + X_n}{n}$
  This would be a reasonable estimate for p, however, there is no way to find the exact value of p on the basis of finite and random poll. Hence, there is going to be an error in the estimation of p.

Revised Problem Definition

- The Boss asks Pollster to calculate the number of people to be randomly picked as a sample of population to estimate the value of the fraction of population that will vote "yes" in a referendum (denoted by 'p') with minimal probability of low accuracy.
- The desired specifications have 2 parameters:
  1. Accuracy of 99% [1% margin of error] and
  2. Probability of an error greater than the margin of error is less than 5% [95% confident with the accuracy that is going to be achieved]

Specifications

$$\Pr\left(|M_n - p| \geq 0.01\right) \leq 0.05$$

Theorem

The Chebyshev's Inequality

1. For a random variable X with finite mean μ and variance σ^2,

$$\Pr\left(|X - \mu| \geq c\right) \leq \frac{\sigma^2}{c^2} \tag{13}$$

   where c is any positive real number.

2. Intuitively, this states that X is unlikely to be too far from the mean if the variance is small.

Solution

Consider the i.i.d random variables X_is that act as an indicator that answer is "yes" in the poll. These follow Bernoulli's Distribution with:

$$\Pr(X_i = x) = \begin{cases} p, & \text{for x} = 1 \\ 1 - p, & \text{for x} = 0 \end{cases} \tag{14}$$

$$\mu = E[X_i] = \sum_{\forall x} \Pr(X_i = x) \times x \tag{15}$$

$$\mu = p \times 1 + (1 - p) \times 0 = p \tag{16}$$

Solution

As X_i is bernoulli random variable, $E[X_i] = E[(X_i)^2] = p$ as x=$\{0,1\}$ is the solution set of $x^2 = x$.

$$\sigma^2 = E[(X_i)^2] - (E[X_i])^2 \tag{17}$$

$$= p - p^2 = p(1-p) \tag{18}$$

Solution

Let M_n be the sample mean of the i.i.d random variables X_is.

$$M_n = \frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} \tag{19}$$

$$E[M_n] = \frac{E[X_1 + X_2 + X_3 + \cdots + X_n]}{n} = \frac{n\mu}{n} = \mu = p \tag{20}$$

$$var(M_n) = \frac{var(X_1 + X_2 + X_3 + \cdots + X_n)}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} = \frac{p(1-p)}{n} \tag{21}$$

Solution - Method 1

By applying Chebyshev's Inequality for M_n, we get:

$$\Pr\left(|M_n - \mu| \geq \epsilon\right) \leq \frac{var(M_n)}{\epsilon^2} = \frac{p(1-p)}{n\epsilon^2} \leq \frac{1}{4n\epsilon^2} \tag{22}$$

Comparing this with the desired specifications, we get:

$$\epsilon = 0.01 \text{ and } \frac{1}{4n\epsilon^2} = 0.05 \tag{23}$$

Solving this, we get: n = 50000

Theorem

The Central Limit Theorem

For a i.i.d random variables $X_1, X_2, \ldots, X_n$ with finite mean μ and variance σ^2, the random variable given by:

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\sigma} \text{ (where: } S_n = X_1 + X_2 + \cdots + X_n) \qquad (24)$$

follows approximately standard normal distribution for larger n.

Solution - Method 2

We have to rewrite the event $|M_n - p| \geq 0.01$ with an equivalent way involving the standardized normal variable Z_n so that we can use Central Limit Theorem to approximate $\Pr(|M_n - p| \geq 0.01)$

$$|M_n - p| \geq 0.01 \implies \left|\frac{S_n}{n} - p\right| \geq 0.01 \tag{25}$$

$$\implies \left|\frac{S_n - np}{n}\right| \geq 0.01 \implies \left|\frac{S_n - np}{\sqrt{n}}\right| \geq 0.01\sqrt{n} \tag{26}$$

$$\implies \left|\frac{S_n - np}{\sqrt{n}\sigma}\right| \geq \frac{0.01\sqrt{n}}{\sigma} \implies |Z_n| \geq \frac{0.01\sqrt{n}}{\sigma} \tag{27}$$

Solution - Method 2

$$\implies |Z_n| \geq \frac{0.01\sqrt{n}}{\sqrt{p(1-p)}} \tag{28}$$

$$\Pr\left(|M_n - p| \geq 0.01\right) \approx \Pr\left(|Z| \geq \frac{0.01\sqrt{n}}{\sqrt{p(1-p)}}\right) \tag{29}$$

$$\Pr\left(|Z| \geq \frac{0.01\sqrt{n}}{\sqrt{p(1-p)}}\right) \leq \Pr\left(|Z| \geq 0.02\sqrt{n}\right) \tag{30}$$

Solution - Method 2

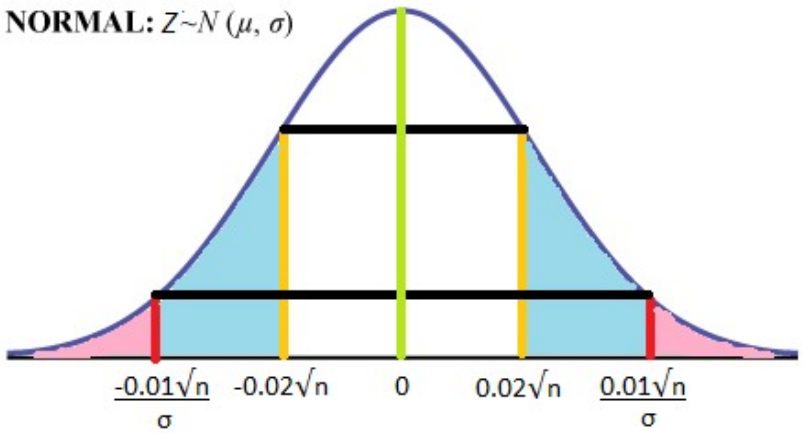


Figure: Normal Distribution of Z

Solution - Method 2

$$\Pr\left(|Z| \geq 0.02\sqrt{n}\right) = 2 \times \Pr\left(Z \geq 0.02\sqrt{n}\right) \tag{31}$$

$$\Pr\left(Z \geq 0.02\sqrt{n}\right) = 1 - \phi(0.02\sqrt{n}) \tag{32}$$

$$\implies \Pr\left(|M_n - p| \geq 0.01\right) \leq 2(1 - \phi(0.02\sqrt{n})) \tag{33}$$

Comparing this with the desired specifications, we get:

$$2(1 - \phi(0.02\sqrt{n})) = 0.05 \tag{34}$$

$$\implies \phi(0.02\sqrt{n}) = 0.975 \tag{35}$$

$$\implies 0.02\sqrt{n} = 1.96 \tag{36}$$

$$\implies n = 9604 \tag{37}$$

Solution - Method 2

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997

Figure: Z Table - Standard Normal Distribution Table

Happy April Fool's Day!



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Any Questions or Suggestions?

Thank You!

Linear Algebra

Linear Independence

- A set of vectors is said to be **linearly independent** if none of the vectors in the set can be expressed as a linear combination of the others.
- A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is linearly independent if the only solution to the equation

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$$

is $a_1 = a_2 = \cdots = a_n = 0$.

Linear Independence

- Consider the following set of vectors in $\mathbb{R}^2$:

$$\{\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}\}$$

  This set is linearly independent as the only solution to the below equation is $a_1 = a_2 = 0$.

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 = \mathbf{0}$$

- In contrast, the following set of vectors in $\mathbb{R}^2$ is linearly dependent as $\mathbf{v}_2$ can be written as a multiple of $\mathbf{v}_1$

$$\{\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}\}$$

Linear Independence

- Consider the following set of vectors:

$$\{\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}\}$$

This set of vectors is linearly independent, since no vector in the set can be expressed as a linear combination of the other two.

Vector spaces

- **Vector space** is a set that contains all the linear combinations of it's vectors.
- A **subspace** is a subset of a vector space.

Span

- The **span** of a set of vectors $x_1, x_2, \ldots, x_n$ is the set of all linear combinations of those vectors and is defined as

$$span(x_1, x_2, \ldots, x_n) = \alpha_1 x_1 + \alpha_2 x_2 + \ldots + \alpha_n x_n \tag{38}$$

- If the vectors $x_1, x_2, \ldots, x_n$ span $\mathbb{R}^n$, then any vector in $\mathbb{R}^n$ can be represented by a linear combination of the vectors $x_1, x_2, \ldots, x_n$.
- The span of any set of vectors in a vector space is a subspace.

Span Example

- Consider 2 vectors $\mathbf{a}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The span of $\mathbf{a}$ and $\mathbf{b}$ is the set of all vectors of the form $c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
- This is the set of all vectors in the 2-D space. Therefore vectors $\mathbf{a}$ and $\mathbf{b}$ spans $\mathbb{R}^2$.

Basis

- A **basis** for a vector space is the minimum set of vectors that spans the sub space or the linearly independent vectors that span the space.
- Examples:
  - From the previous example, the set of vectors $\mathbf{a}$ and $\mathbf{b}$ is a basis for the 2-dimensional Euclidean space.
  - The standard basis for $\mathbb{R}^n$ is $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$, where $\mathbf{e}_i$ is the i-th canonical basis vector.

Basis

- **Theorem:** Every finite-dimensional vector space has a basis.
  - Example: Find a basis for the vector space spanned by $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix}$.
  - Solution: Since the two vectors are linearly dependent, we can remove one of them to get a basis. For example, $\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}$ is a basis for the given vector space.

Linear Transformations

- A linear transformation T is a function that maps vectors from one vector space to another, while preserving the structure of the vector space.
- If v_1 and v_2 are 2 vectors in the input space, and c is a scalar, then the mapping T:V $\longrightarrow$ W is a linear transformation when
  - $T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2)$ or $T(v_1 + v_2) = T(v_1) + T(v_2)$ and
  - $T(cv_1) = cT(v_1)$

Linear Transformations

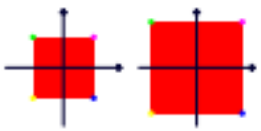
- In terms of matrix multiplication, when we multiply an $m \times n$ matrix by an $n \times 1$ column vector, the result is an $m \times 1$ column vector. The notation T: $\mathbb{R}^n \longrightarrow \mathbb{R}^m$ is transformation from an m-dimensional vector space to n-dimensional vector space.
- A is an $m \times n$ matrix with column vectors $\mathbf{v_1}, \mathbf{v_2}, \cdots, \mathbf{v_n}$.

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} \mathbf{v_1} & \mathbf{v_2} & \cdots & \mathbf{v_n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{v_1} + x_2\mathbf{v_2} + \cdots + x_n\mathbf{v_n}.$$

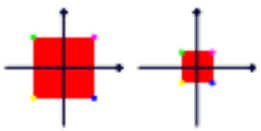
Types of Linear Transformations

- **Scaling Transformation** is a linear transformation that scales a vector by a constant factor and is represented by a diagonal matrix.

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

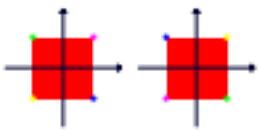


$$A = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

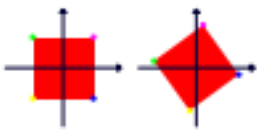


- **Rotation Transformation** is a linear transformation that rotates a vector by a certain angle around the origin and is represented by a rotation matrix.

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$



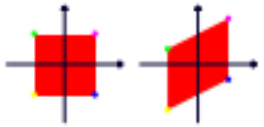
$$A \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix} =$$



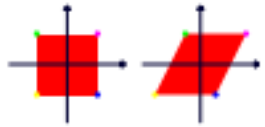
Types of Linear Transformations

- **Shear Transformation** is a linear transformation that shears a vector along one axis.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

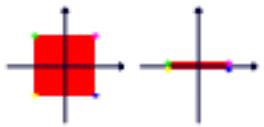


$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

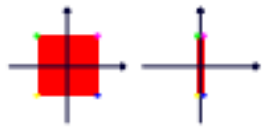


- **Projection Transformation** is a linear transformation that projects a vector onto a lower-dimensional subspace.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$



$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$



The Four Fundamental Subspaces

- The column space $C(A)$
- The row space $C(A^T)$
- The null space $N(A)$
- The null space of the transpose $N(A^T)$

Column Space and Row Space

- The **column space** of a matrix is the space spanned by its column vectors.
- Column space of a matrix $\mathbf{A}$ is denoted by $\mathrm{col}(\mathbf{A})$ or $C(A)$.

$$\mathrm{col}(\mathbf{A}) = \mathrm{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$$

  Here $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n \in \mathbb{R}^m$ and are the column vectors of $\mathbf{A}$.
- The **row space** of matrix A is the space spanned by the rows vectors of A. It is denoted by $C(A^T)$.

Null Space

- The **null space** of a matrix A, denoted as $N(A)$, is the set of all solutions of $\mathbf{x}$ to the homogeneous equation $Ax = 0$.
- $N(A) = \{ \mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = 0 \}$.

Dimension

- The **dimension** of a subspace is the number of elements in a basis for the sub space.
- The dimension of $N(A)$ is the **nullity of** A and is denoted by $nullity(A)$.
- The dimension of the column space is called the rank of **A**.
- The column space and row space have the **same dimension** and is equal to the **rank**.

Row operations

- **Row operations** are manipulating the rows of a matrix.
- There are three types of row operations: **swapping two rows**, **multiplying a row by a scalar**, and **adding a multiple of one row to another row**.
- Row operations are used to solve systems of linear equations.

Example

- Let's consider the following system of linear equations:

$$\begin{aligned} 2x + 3y - z &= 1 \\ 4x + 5y - 3z &= 2 \\ 2x + y + z &= 0 \end{aligned}$$

Augmented matrix:

$$\left[\begin{array}{ccc|c} 2 & 3 & -1 & 1 \\ 4 & 5 & -3 & 2 \\ 2 & 1 & 1 & 0 \end{array}\right]$$

Perform a series of elementary row operations on the augmented matrix to simplify it.

Rank

- The rank of a matrix is defined as the **maximum number of linearly independent rows or columns** in the matrix.
- The dimension of the column space is called the rank of **A**.

Row reduction

Row reduction is equivalent to left-multiplying the matrix by another invertible matrix E. This means that the net effect of making a row reduction is basically $=$

$$CA = R$$

where C is an invertible square matrix.

Standard Results

- Independent Cols $\implies Ax = 0 \implies x = 0$

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- LI Columns means Rows must span $\mathbb{R}^m$
- LI rows means Columns span $\mathbb{R}^n$
- LI rows means the mapping is onto.

Rank-Nullity theorem

If A is $m \times n$

Then, the Dimension of the column space + the dimension of the null space = n

$Ax = b$

Every solution for $Ax = b$ is of the form $x_p + x_n$

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Why tho?

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Why tho?
Proof by contradiction

Finding the Solution

To find a solution for $Ax = b$, we first check if b belongs to the column space for A. We will construct the augmented matrix $[A\ b]$ and perform row reduction.

The Four fundamental Subspaces

The four fundamental subspaces have a very nice little structure.

- $C(A^T)$(r) and $N(A)$(n - r) are orthogonal to each other
- $C(A)$(r) and $N(A^T)$(m - r) are orthogonal to each other
- These spaces are called **orthogonal Complements** of each other.

Direct Sum of the spaces

- The direct Sum of two Subspaces is basically the set of vectors v such that it can be written as $u + v$ where $u \in S$ and $v \in S^{\perp}$

Projections

Say that We are attempting to solve the problem of projecting the vector b onto the columns space of some matrix A, i.e we are trying to solve for the value of x that yields

$$A\hat{x} \approx b$$

Then the projected value of b on the column space would be something like $A\hat{x} = AA^{+}b$. But does this maintain the "perpendicular-ness" property of projections?

Projections

We simply apply the property that perpendicular vectors have a zero dot product and we solve for the value of x.

$$A^T(b - Ax) = 0 \tag{39}$$

$$A^T b - A^T Ax = 0 \tag{40}$$

$$x = (A^T A)^{-1} A^T b = A^+ b \tag{41}$$

QR Decomposition

What do I do if I do not have a basis for the column space? How do I find the projection matrix then?

- Remove those columns that are dependent and then continue
- Be smarter, and do a QR decomposition to find an **orthonormal basis** for the column space.

QR

$$\mathbf{A}_{m\times n} = \mathbf{Q}_{m\times m}\mathbf{R}_{m\times n}. \tag{22}$$

Let us take as following.

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_{:1} & \mathbf{a}_{:2} & \dots & \mathbf{a}_{:n} \end{bmatrix}, \tag{23}$$

$$\mathbf{Q} = \begin{bmatrix} \mathbf{e}_{:1} & \mathbf{e}_{:2} & \dots & \mathbf{e}_{:m} \end{bmatrix}. \tag{24}$$

The process is as follows.

$$\mathbf{u_1} = \mathbf{a}_{:1}, \qquad \mathbf{e}_{:1} = \frac{\mathbf{u_1}}{\|\mathbf{u_1}\|}, \tag{25}$$

$$\mathbf{u_2} = \mathbf{a}_{:2} - \langle \mathbf{a}_{:2}, \mathbf{e}_{:1} \rangle \mathbf{e}_{:1}, \qquad \mathbf{e}_{:2} = \frac{\mathbf{u_2}}{\|\mathbf{u_2}\|}, \tag{26}$$

$$\mathbf{u_3} = \mathbf{a}_{:3} - \langle \mathbf{a}_{:3}, \mathbf{e}_{:1} \rangle \mathbf{e}_{:1} - \langle \mathbf{a}_{:3}, \mathbf{e}_{:2} \rangle \mathbf{e}_{:2}, \qquad \mathbf{e}_{:3} = \frac{\mathbf{u_3}}{\|\mathbf{u_3}\|}, \tag{27}$$

$$\therefore \mathbf{u_{k+1}} = \mathbf{a}_{:k+1} - \sum_{i=1}^{k} \langle \mathbf{a}_{:k+1}, \mathbf{e}_{:i} \rangle, \qquad \mathbf{e}_{:k+1} = \frac{\mathbf{u_{k+1}}}{\|\mathbf{u_{k+1}}\|}. \tag{28}$$

$$\tag{29}$$

Hence, we get the values of $\mathbf{Q}$, now for $\mathbf{R}$, we need to write $\mathbf{a}_{:i}$ as a linear combination of $\mathbf{e}_{:j}$.

$$\mathbf{a}_{:1} = \langle \mathbf{a}_{:1}, \mathbf{e}_{:1} \rangle \mathbf{e}_{:1}, \tag{30}$$

$$\mathbf{a}_{:2} = \langle \mathbf{a}_{:2}, \mathbf{e}_{:1} \rangle \mathbf{e}_{:1} + \langle \mathbf{a}_{:2}, \mathbf{e}_{:2} \rangle \mathbf{e}_{:2}, \tag{31}$$

$$\mathbf{a}_{:3} = \langle \mathbf{a}_{:3}, \mathbf{e}_{:1} \rangle \mathbf{e}_{:1} + \langle \mathbf{a}_{:3}, \mathbf{e}_{:2} \rangle \mathbf{e}_{:2} + \langle \mathbf{a}_{:3}, \mathbf{e}_{:3} \rangle \mathbf{e}_{:3}, \tag{32}$$

$$\therefore \mathbf{a}_{:i} = \sum_{j=1}^{i} \langle \mathbf{a}_{:i}, \mathbf{e}_{:j} \rangle \mathbf{e}_{:j}. \tag{33}$$

$\mathbf{R}$ matrix will consist of ith column as the coefficients of each of $\mathbf{e}_{:j}$ that sum up to $\mathbf{a}_{:i}$.

GSO

The GSO Procedure is simply constructing an orthonormal basis using a set of "non-orthogonal" vectors. The set of equations can be arranged in the following format:-

SVD

Find the vectors that maximize the value of the following terms, successively.

$$\max_{v_i \perp v_j \forall j<i \ \|v_i\|=1} \|A\mathbf{v_i}\| = \sigma_i u_i$$

- The vectors v are orthogonal to each other(obviously)

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Singular Value Decomposition

A transform that is described by a Matrix can be seen entirely as a bunch of "Scaling" and Orthogonal Transformations.

Full SVD:

Every Matrix $A \in \mathbb{R}^{m \times n}$ can be broken down into three Matrices U, Σ, and V such that

$$A = U \Sigma V^* \tag{42}$$

with $U \in \mathbb{R}^{m \times m}, \Sigma \in \mathbb{R}^{m \times n}, V \in \mathbb{R}^{n \times n}$

Fun Results on SVD

- $Col(A) = Col(U_1)$
- $Col(A^T) = Col(V_1)$

Full SVD using Basis Extension Just extend the Basis for U_1, V_1 and add a bunch of zeroes to the diagonal matrix Σ. Then we can also find the basis for the Nullspaces of A

$$\boxed{\mathbf{A} = [\mathbf{U_1} \quad \mathbf{U_2}]_{m\times m} \begin{bmatrix} \mathbf{\Sigma_1} & 0 \\ 0 & 0 \end{bmatrix}_{m\times n} \begin{bmatrix} \mathbf{V_1^T} \\ \mathbf{V_2^T} \end{bmatrix}_{n\times n}}.$$

Eigenvalues

The eigenvectors represent an invariant subspace of the matrix A

$$Ax = \lambda x \tag{43}$$
$$(A - \lambda I)x = 0 \tag{44}$$
$$det(A - \lambda I) = 0 \tag{45}$$

Every Eigenvalue is a root of the equation [45]. Also, note that this is a polynomial equation in λ, FTA tells us that this must have n roots exactly

Results on Eigenvalues

Some easy results to show (in the exercises):-

Some Results

$$\sum \lambda_i = \mathrm{Tr}(A)$$
$$\prod \lambda_i = \det(A)$$
$$Eig(A + \mu I) = Eig(A) + \mu$$

Eigenvalues can occur with a certain algebraic multiplicity

Similarity

Two matrices are said to be similar if there exists a non singular S such that

$$B = S^{-1}AS \implies B \sim A \quad \text{(It is an Equivalence Relation :)}$$

The meaning is that B is A after we transform the space into the Column space of S. Since S is invertible and Square, we can see that we aren't *squishing* the space.

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CP

The CP of two similar matrices are the same, i.e., you can see that similar matrices also have equal eigenvalues.

Diagonalizability

A Matrix is said to be diagonalizable, if it is similar to a diagonal matrix. Now, we can analyze when exactly matrices are diagonalizable?

$$AS = S\Lambda \implies As_i = \lambda_i s_i$$

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$$AS = S\Lambda \implies As_i = \lambda_i s_i$$

It turns out the columns of S must form a square matrix and the columns of S must become the eigenvectors of this matrix A

Normal Matrices

- Normal Matrices are Diagonalizable.
- Tf is Normal? and Why are they diagonalizable?
- Normal Matrices are A such that $AA^* = A^*A$

Quadratic Forms

Analyzing the expressions of the form $x^T Ax$, when the norm of x is bounded

$$x^T Ax = x^T S \Lambda S^T x = y^T \Lambda y = \sum_i \lambda_i y_i^2$$

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PSD Matrices

PSD matrices are those matrices for which their quadratic forms are always nonegative. Can you see why this would mean that all their eigenvalues must also be non-negative?

More Advanced Theorems

Calculus

Introduction

Calculus is an important tool in machine learning. In this session, we will cover the basics of numerical differentiation, numerical integration, and gradient descent, and how they are used in machine learning.

Numerical Differentiation

Numerical differentiation is the process of approximating the derivative of a function at a point using finite differences. There are several methods for numerical differentiation, including forward difference, backward difference, and central difference.

Forward Difference Method

The forward difference method approximates the derivative using the formula:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

where h is a small step size.

Backward Difference Method

The backward difference method approximates the derivative using the formula:

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

where h is a small step size.

Central Difference Method

The central difference method approximates the derivative using the formula:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

where h is a small step size.

Difference between the methods

- Forward difference is useful when we need to approximate the derivative at a point using only the data points that come after it.
- While backward difference is useful when we need to use only the data points that come before the point of interest.
- Central difference, on the other hand, uses data points before and after the point of interest, and is useful when we need a more accurate estimate of the derivative.

Other Techniques

- Richardson extrapolation: This method uses a combination of approximations obtained using different step sizes to obtain a more accurate estimate of the derivative.
- Spectral methods: These are numerical techniques that use Fourier series or other orthogonal functions to approximate the function and its derivatives.
- Automatic differentiation: This method uses computer algorithms to automatically calculate the derivative of a function, without the need for numerical approximations.

Numerical Integration

- Bayesian inference: In Bayesian inference, we often need to compute integrals over posterior distributions, which can be difficult or impossible to do analytically. Numerical integration methods, such as Markov chain Monte Carlo (MCMC) and variational inference, are commonly used to approximate these integrals.
- Reinforcement learning: In reinforcement learning, the expected value of a reward function is often computed using numerical integration methods. For example, in the Monte Carlo policy evaluation algorithm, we approximate the expected return by averaging the rewards obtained from multiple episodes of interaction with the environment.

Numerical Integration

- Neural networks: In deep learning, numerical integration can be used to compute the expected value of the loss function over a batch of training data. This is done using methods such as stochastic gradient descent (SGD) and its variants, which approximate the integral by randomly sampling mini-batches of data.
- Gaussian processes: Gaussian processes are a type of probabilistic model commonly used in machine learning. Inference in Gaussian processes involves computing integrals over Gaussian distributions, which can be done using numerical integration methods such as quadrature and Monte Carlo integration.

Trapezoidal Method

The trapezoidal rule approximates the definite integral using the formula:

$$\int_a^b f(x)dx \approx \frac{b-a}{2n}[f(a)+2f(a+h)+2f(a+2h)+.....2f(b-h)+f(b)]$$

where, $h = \frac{b-a}{n}$ and n is the number of intervals.

Simpson's Method

Simpson's rule approximates the definite integral using the formula:

$$\int_a^b f(x)dx \approx \frac{b-a}{6n}[f(a) + 4f(a+h) + 2f(a+2h) + 4f(a+3h) +$$

$$....2f(b-2h) + 4f(b-h) + f(b)]$$

where $h = \frac{b-a}{2n}$ and n is the number of intervals

Gradient Descent

Gradient descent is an optimization algorithm used in machine learning to minimize a loss function. The goal is to find the values of the parameters that minimize the loss function. Gradient descent works by iteratively adjusting the parameters in the direction of the negative gradient of the loss function.

Gradient Descent Algorithm

1. Start with an initial guess for the minimum of the function.
2. Compute the gradient
3. Update the parameters using the following rule

$$x = x - \alpha \frac{dy}{dx}$$

4. Take a step in this direction.
5. Repeat steps 2-4 until convergence to a minimum is achieved.

Conclusion

Numerical methods play a critical role in calculus for machine learning algorithms. In this report, we have discussed the numerical methods used for differentiation, integration and optimization. These methods are essential tools for solving mathematical problems that arise in machine learning. By using numerical methods, we can approximate complex functions, solve optimization problems, and model dynamic systems.

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Any Questions or Suggestions?

Thank You!